



The output is high. So, NAND gate is a answer.

6. (c) $\sqrt{(224)}_5 = (13)_5$,

On squaring both sides,

$$(224)_5 = (13)_5^2,$$

Now, converting to decimal number

$$2r^2 + 2r + 4 = (1r + 3)^2$$

$$2x^2 + 2r + 4 = (r + 3)^2$$

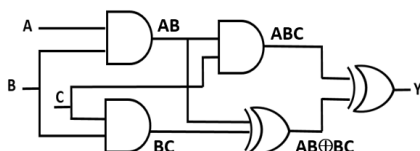
$$2x^2 + 2r + 4 = r^2 + 9 + 6r$$

$$r^2 - 4r - 5 = 0$$

Root of the above equation is 5, -1

Therefore, radix = 5

7. (c)



$$\text{So, } Y = ABC \oplus (AB \oplus BC)$$

$$Y = B(C + A)$$

8. (d)

The exclusive NOR and exclusive OR of x and y is given by the expressions.

$$x \odot y = xy + x'y' \text{ (exclusive NOR or XNOR)}$$

$$x \oplus y = x'y + xy' \text{ (exclusive OR or XOR)}$$

So, option (A) is automatically true.

Option (B) is $x \oplus y'$ which evaluates to $x'y' + (x)(y')$

$$= x'y' + xy \text{ which is equivalent to } x \odot y$$

Option (C) is $x' \oplus y$ which evaluates to

$$(x')y + x'y' = xy + x'y' \text{ which is equivalent to } x \odot y$$

Option (D) is $x' \oplus y'$ which evaluates to $(x')y' + x'(y)'$

$$= xy' + x'y \text{ which is not equivalent to } x \odot y$$

9. (c)

Logical error.

10. (b)

$$(43)_x = (y3)_8$$

Since, a number in base K can only have digits from 0 to $(K - 1)$ we can conclude that

$$x \geq 5 \text{ and } y \leq 7$$

Now, the original equation, when converted to decimal base gives:

$$4x^1 + 3x^0 = y(8^1) + 3(8^0)$$

$$\Rightarrow 4x + 3 = 8y + 3$$

$$\Rightarrow x = 2y$$

So, we have the following constraints

$$x \geq 5$$

$$y \leq 7$$

$x = 2y$, x and y are integers.

The set of values of (x, y) that satisfy these constraints are:

$$(6, 3), (8, 4), (10, 5), (12, 6), (14, 7)$$

So, the number of possible solution is 5.

